

Mihir Mohite

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PROFESSIONAL SUMMARY

A results-driven B.Tech student specializing in AI & ML with extensive hands-on experience in Large Language Models, Retrieval-Augmented Generation (RAG), and agentic AI systems. Currently serving as President of CoDeC and Machine Learning Intern at BeyondBot Technology, with a proven track record of architecting production-grade ML solutions including enterprise knowledge graphs, multi-modal document processing pipelines, and scalable cloud-native applications. Expertise spans the full ML lifecycle from research to deployment, with strong foundation in modern AI frameworks, cloud infrastructure, and full-stack development.

EDUCATION

MIT-WPU, Pune

Expected May 2027

B.Tech in Electronics & Communication Engineering, Specialization in AI & ML

CGPA: 8.5/4.00

Kalmadi Shamrao Junior College, Pune

June 2020 - May 2022

Higher Secondary Certificate, Science

Millennium National School, Pune

Graduated March 2020

CBSE

EXPERIENCE

Machine Learning Intern, BeyondBot Technology Pvt. Ltd., Pune

Oct 2025 - Present

- Developing and optimizing Large Language Model (LLM) based solutions for production environments with focus on performance and scalability.
- Implementing Computer Vision and OCR pipelines for multi-format document processing and information extraction.
- Architecting Retrieval-Augmented Generation (RAG) and GraphRAG systems for enhanced information retrieval and contextual understanding.
- Leveraging cutting-edge ML frameworks and cloud infrastructure to build robust, enterprise-grade AI applications.

Software Intern, Heera Software Private Limited, Pune

June 2025 - Oct 2025

- Developed an address validation system using a custom ML-based multi-class classification model.
- Contributed to production software systems with focus on data quality and validation pipelines.

LEADERSHIP & EXTRACURRICULARS

President, CoDeC - Competitive Programming Club at MITWPU

Oct 2025 - Present

- Leading the strategic direction and technical initiatives for university's premier programming club.
- Organizing workshops, hackathons, and competitive programming events to foster coding excellence.
- Mentoring students in algorithms, data structures, and software engineering best practices.

ML Projects Division Lead, CoDeC

Jan 2025 - Present

- Spearheading AI/ML project initiatives from conception to completion, defining project scope and technology stack.
- Supervising development teams and conducting knowledge-sharing sessions on deep learning frameworks and data science.

SELECTED PROJECTS

University Feedback System

- Developed university-wide feedback platform with WhatsApp integration using Meta Business API for seamless user experience.
- Implemented agentic AI workflow using Agno framework for dynamic question generation and adaptive surveying.

- Built serverless architecture on AWS (Lambda, S3) with React.js frontend for scalable, real-time feedback collection.
- Integrated Vertex AI API with Gemini models (2.0 Flash, 2.5 Flash, 2.5 Pro) for sentiment analysis and feedback summarization.
- Acknowledged by university administration for innovative approach to student engagement and feedback analysis.

EKG - Enterprise Knowledge Graph

Nov 2025

- Engineered enterprise knowledge graph generation system supporting structured (SQL), semi-structured (JSON), and unstructured (TXT) data sources.
- Implemented Neo4j graph database with Gemini API for intelligent entity extraction and relationship mapping.
- Developed interactive dashboard for graph visualization and natural language querying against enterprise data.
- Designed proof-of-concept with scalable architecture demonstrating significant potential for production deployment across industries.

Advanced RAG Implementation

- Built production-grade RAG system using Sentence Transformers for tokenization, chunking, and embedding with Pinecone vector database.
- Implemented alternative RAG pipeline using IBM Docling with Hybrid, Base, and Hierarchical chunking strategies for optimized retrieval.
- Developed multi-format document processing supporting PDF, JPG, XLSX, DOCX with local Milvus client for vector storage.
- Maximized RAG capabilities through systematic experimentation with chunking strategies and embedding techniques.

Scout Rover for Dangerous Environments

- Built autonomous rover with real-time hazardous area mapping using LiDAR-based SLAM in MATLAB.
- Integrated gas, temperature, and humidity sensors for comprehensive environmental monitoring.

Electric Load Forecasting (Delhi)

- Designed deep-learning MLP model using Python and TensorFlow/Keras for short-term and long-term hourly electricity demand prediction.
- Utilized time-series analysis to process historical load data for accurate forecasting.

CommonForms - Form Field Detection

- Trained CNN on Common Crawl dataset using FFDNet (Form Field Detection Network) for automated form field detection.
- Developed robust classification system for identifying and extracting form fields from diverse document types.

TECHNICAL SKILLS

AI/ML & LLMs: RAG (Sentence Transformers, Pinecone, Milvus), IBM Docling (Hybrid/Hierarchical Chunking), Vertex AI, Gemini API (2.0 Flash, 2.5 Flash, 2.5 Pro), Agno Agentic Framework, NLP, Computer Vision, OCR, PyTorch, TensorFlow, Keras, scikit-learn, Transformer Architecture, Vector Embeddings, FFDNet

Web Development: React.js, Node.js, HTML5, CSS3, JavaScript, SQL

Cloud & DevOps: AWS (Lambda, S3, EC2, Transcribe, Secrets Manager), Google Cloud (Vertex AI)

Databases & Knowledge Graphs: Neo4j, Pinecone, Milvus, SQL, MySQL

Programming Languages: Python, C, C++, Java, Embedded C

Tools & Frameworks: Git, MATLAB, PyMuPDF, SLAM, FFT, Algorithm Design, WhatsApp Business API, Meta API

Hardware & IoT: Arduino, ESP32/8266, Raspberry Pi, LiDAR, LoRa, Circuit Design

ACHIEVEMENTS

- Smart India Hackathon (SIH) Round 2 Qualifier
- Runner-up, Circuit Heist Hackathon
- Final Rounds Qualifier, Dataquest AI Challenge, HackMITWPU